

AI Assisted Coding

Lab Assignment - 14.5

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Lab 14: Web Design Application – AI-Assisted HTML/CSS/JS Generation

Lab Objectives

- Design functional, visually appealing web applications using HTML, CSS, and JavaScript with AI assistance.
- Apply responsive, accessible, and interactive design principles.
- Create practical UI components for real-world web applications.
- Use AI to optimize layout, UX, and performance.

Task 1 – Image Gallery Website

Create an image gallery web page.

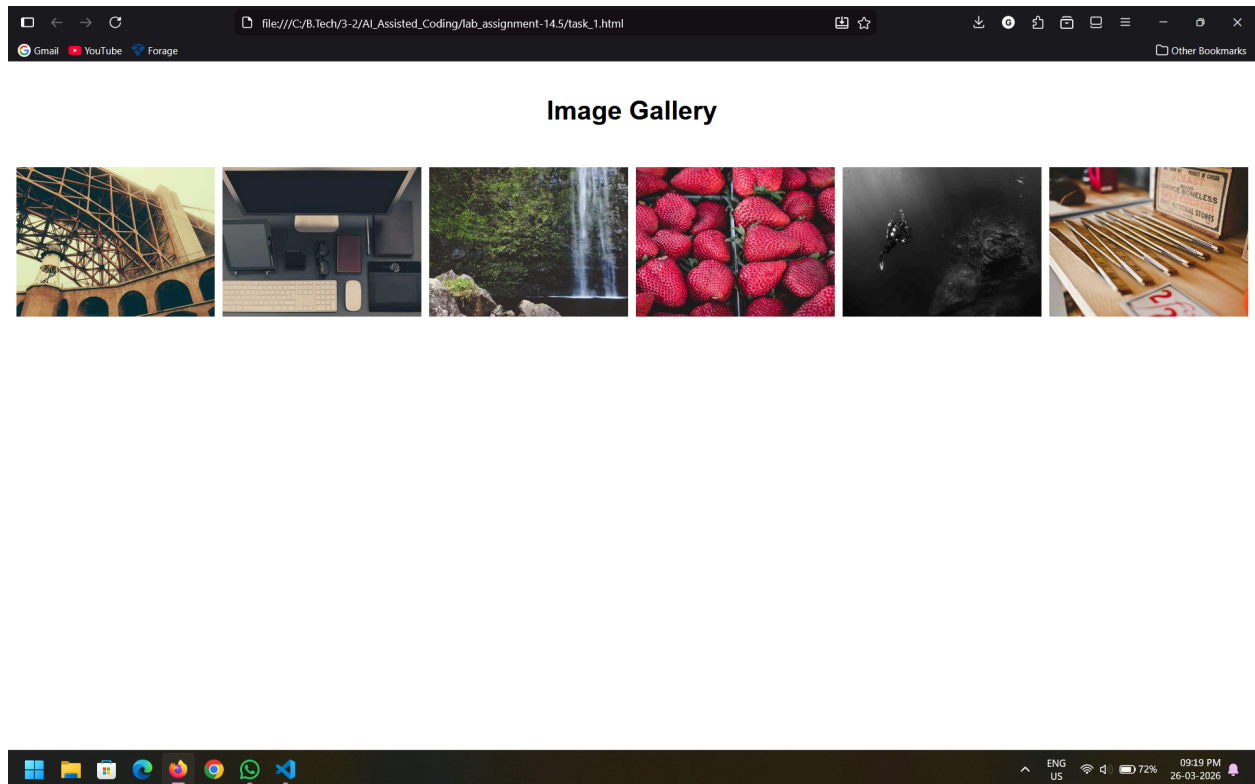
Requirements:

- Display images in a grid layout.
- Use AI to:
 - o Optimize layout using CSS Grid.
 - o Add hover zoom or overlay effects.
 - o Ensure mobile responsiveness.

Copilot Suggested Code :

```
task_1.html > html
1  <!-- Create an image gallery web page. -->
2
3  <!DOCTYPE html>
4  <html lang="en">
5  <head>
6      <title>Image Gallery</title>
7      <style>
8          body {
9              font-family: 'Arial', sans-serif;
10             margin: 0;
11             padding: 0;
12         }
13         .gallery {
14             display: grid;
15             grid-template-columns: repeat(auto-fit, minmax(200px, 1fr));
16             gap: 10px;
17             padding: 10px;
18         }
19         .gallery img {
20             width: 100%;
21             height: auto;
22             transition: transform 0.3s ease, box-shadow 0.3s ease;
23         }
24         .gallery img:hover {
25             transform: scale(1.05);
26             box-shadow: 0 4px 8px rgba(0,0,0,0.2);
27         }
28         @media (max-width: 600px) {
29             .gallery {
30                 grid-template-columns: repeat(auto-fit, minmax(150px, 1fr));
31             }
32         }
33     </style>
34 </head>
35 <body>
36     <h1 style="text-align: center; padding: 20px;">Image Gallery</h1>
37     <div class="gallery">
38         
39         
40         
41         
42         
43         
44     </div>
45 </body>
46 </html>
```

Output :



Task 2 – Profile Card Generator

Create a profile card web page.

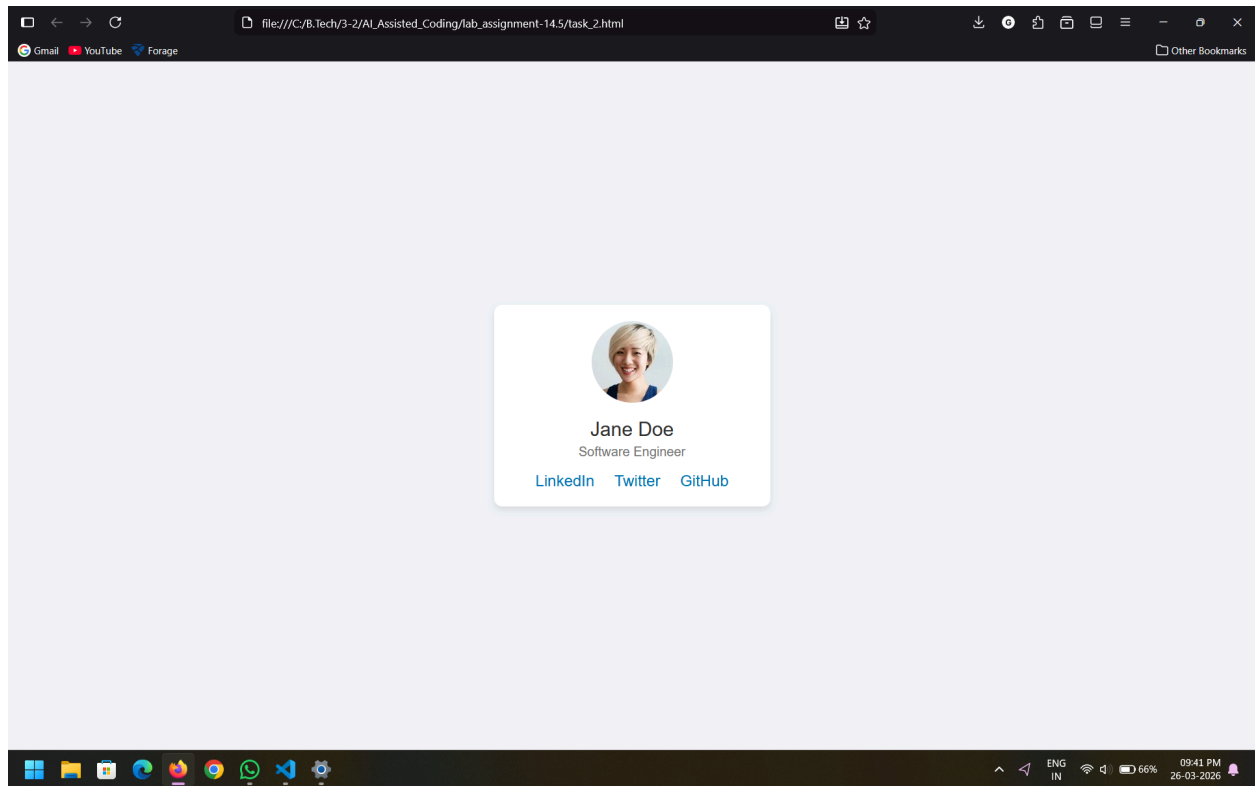
Requirements:

- Display user image, name, role, and social media links.
- Use AI to:
 - o Suggest modern UI colors and fonts.
 - o Design the card using Flexbox.
 - o Add hover animation and shadow effects

Copilot Suggested Code :

```
task_2.html > html
1  <!-- Create a profile card web page. -->
2
3  <!DOCTYPE html>
4  <html lang="en">
5  <head>
6      <title>Profile Card</title>
7      <style>
8          body {
9              font-family: 'Helvetica Neue', sans-serif;
10             background-color: #f0f4f8;
11             display: flex;
12             justify-content: center;
13             align-items: center;
14             height: 100vh;
15             margin: 0;
16         }
17         .profile-card {
18             background-color: #ffffff;
19             border-radius: 10px;
20             box-shadow: 0 4px 8px rgba(0,0,0,0.1);
21             width: 300px;
22             padding: 20px;
23             text-align: center;
24             transition: transform 0.3s ease, box-shadow 0.3s ease;
25         }
26         .profile-card:hover {
27             transform: translateY(-5px);
28             box-shadow: 0 8px 16px rgba(0,0,0,0.2);
29         }
30         .profile-image {
31             width: 100px;
32             height: 100px;
33             border-radius: 50%;
34             object-fit: cover;
35             margin-bottom: 15px;
36         }
37         .profile-name {
38             font-size: 1.5em;
39             color: #333333;
40             margin-bottom: 5px;
41         }
42         .profile-role {
43             font-size: 1em;
44             color: #777777;
45             margin-bottom: 15px;
46         }
47         .social-links a {
48             color: #0077b5; /* LinkedIn Blue */
49             text-decoration: none;
50             margin: 0 10px;
51             font-size: 1.2em;
52         }
53     </style>
54 </head>
55 <body>
56     <div class="profile-card">
57         
58         <div class="profile-name">Jane Doe</div>
59         <div class="profile-role">Software Engineer</div>
60         <div class="social-links">
61             <a href="https://linkedin.com/in/jane-doe" target="_blank">LinkedIn</a>
62             <a href="https://twitter.com/jane_doe" target="_blank">Twitter</a>
63             <a href="https://github.com/jane-doe" target="_blank">GitHub</a>
64         </div>
65     </div>
66 </body>
67 </html>
```

Output :



Task 3 -Develop a dynamic shopping cart system.

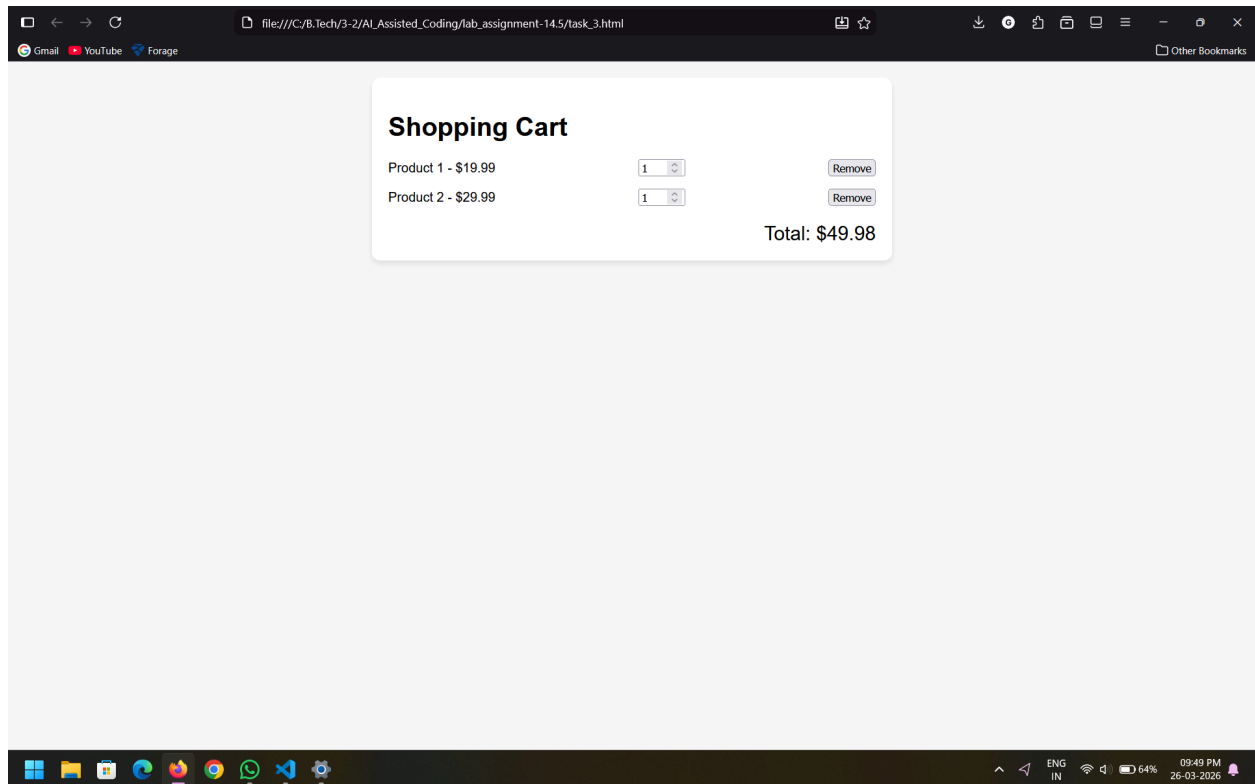
Requirements:

- Add, remove, and update product quantities.
- Display real-time total price.
- Use AI to:
 - o Manage cart state using JavaScript objects.
 - o Persist cart data in localStorage.

Copilot Suggested Code :

```
1  <!-- Task 3 -Develop a dynamic shopping cart system.
2  Requirements:
3  * Add, remove, and update product quantities.
4  * Display real-time total price.
5  * Use AI to:
6  o Manage cart state using JavaScript objects.
7  o Persist cart data in localStorage. -->
8
9  <!DOCTYPE html>
10 <html lang="en">
11 <head>
12   <title>Shopping Cart</title>
13   <style>
14     body {
15       font-family: 'Arial', sans-serif;
16       margin: 0;
17       padding: 20px;
18       background-color: #f9f9f9;
19     }
20     .cart-container {
21       max-width: 600px;
22       margin: 0 auto;
23       background-color: #fff;
24       padding: 20px;
25       border-radius: 10px;
26       box-shadow: 0 4px 8px rgba(0,0,0,0.1);
27     }
28     .cart-item {
29       display: flex;
30       justify-content: space-between;
31       align-items: center;
32       margin-bottom: 15px;
33     }
34     .cart-item input {
35       width: 50px;
36     }
37     .total-price {
38       font-size: 1.5em;
39       text-align: right;
40       margin-top: 20px;
41     }
42   </style>
43 </head>
44 <body>
45   <div class="cart-container">
46     <h1>Shopping Cart</h1>
47     <div id="cart-items"></div>
48     <div class="total-price" id="total-price">Total: $0.00</div>
49   </div>
50
51   <script>
52     const cart = JSON.parse(localStorage.getItem('cart')) || {};
53
54     function updateCart() {
55       const cartItemsContainer = document.getElementById('cart-items');
56       cartItemsContainer.innerHTML = '';
57       let totalPrice = 0;
58
59       for (const productId in cart) {
60         const item = cart[productId];
61         totalPrice += item.price * item.quantity;
62
63         const cartItem = document.createElement('div');
64         cartItem.className = 'cart-item';
65         cartItem.innerHTML = `
66           <span>${item.name} - ${item.price.toFixed(2)}</span>
67           <input type="number" value="${item.quantity}" min="1" onchange="updateQuantity('${productId}', this.value)">
68           <button onclick="removeFromCart('${productId}')">Remove</button>
69         `;
70         cartItemsContainer.appendChild(cartItem);
71       }
72
73       document.getElementById('total-price').textContent = `Total: ${totalPrice.toFixed(2)}`;
74       localStorage.setItem('cart', JSON.stringify(cart));
75     }
76
77     function addToCart(product) {
78       if (cart[product.id]) {
79         cart[product.id].quantity += 1;
80       } else {
81         cart[product.id] = { ...product, quantity: 1 };
82       }
83       updateCart();
84     }
85
86     function updateQuantity(productId, quantity) {
87       if (cart[productId]) {
88         cart[productId].quantity = parseInt(quantity);
89         updateCart();
90       }
91     }
92
93     function removeFromCart(productId) {
94       delete cart[productId];
95       updateCart();
96     }
97
98     // Example products
99     addToCart({ id: '1', name: 'Product 1', price: 19.99 });
100    addToCart({ id: '2', name: 'Product 2', price: 29.99 });
101  </script>
102 </body>
103 </html>
```

Output :



Task 4 – Role-Based Access Control Application

Create a role-based web application.

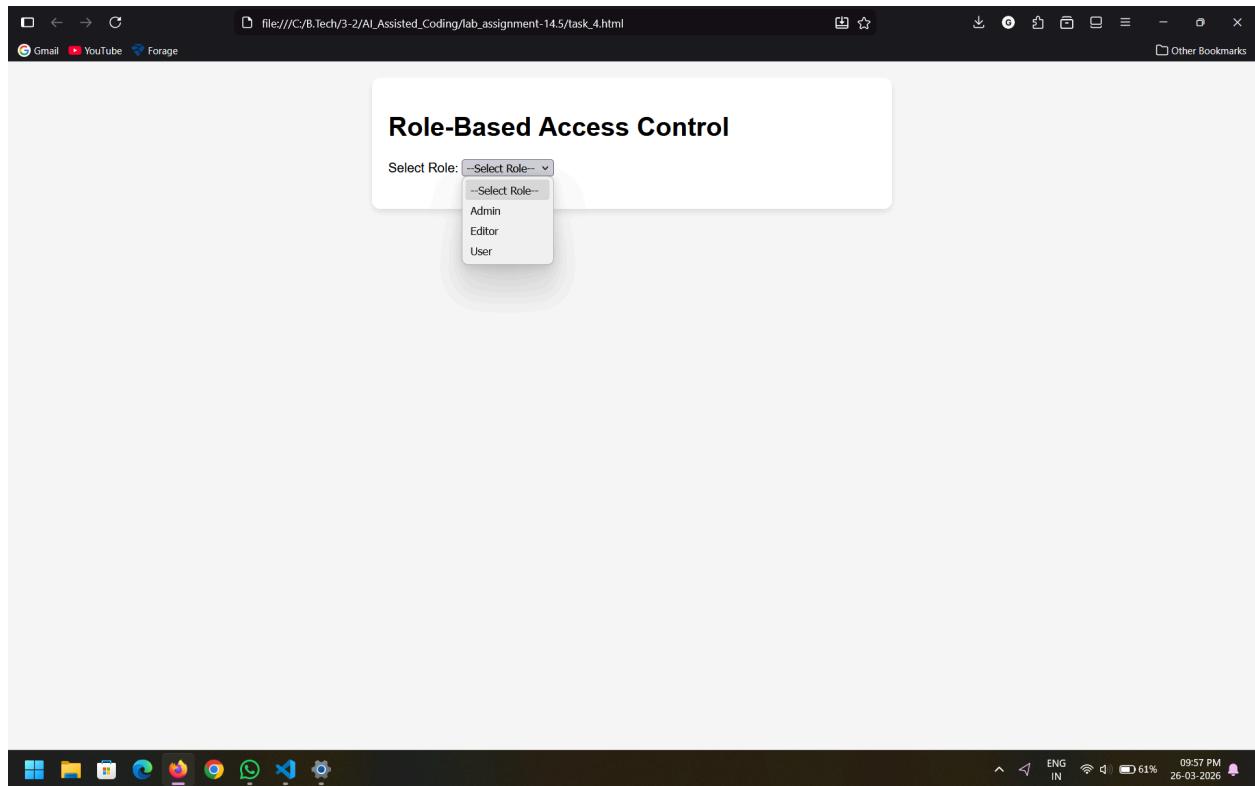
Requirements:

- Admin, Editor, User roles.
- Conditional UI rendering.
- Use AI to:
 - o Design access control logic.
 - o Maintain secure state transitions.
 - o Create scalable UI components.

Copilot Suggested Code :

```
task_4.html > ...
1  <!-- create a role-based web application. -->
2
3  <!DOCTYPE html>
4  <html lang="en">
5  <head>
6    <title>Role-Based Access Control</title>
7    <style>
8      body {
9        font-family: 'Arial', sans-serif;
10       margin: 0;
11       padding: 20px;
12       background-color: #f9f9f9;
13     }
14     .container {
15       max-width: 600px;
16       margin: 0 auto;
17       background-color: #fff;
18       padding: 20px;
19       border-radius: 10px;
20       box-shadow: 0 4px 8px rgba(0,0,0,0.1);
21     }
22     .role-select {
23       margin-bottom: 20px;
24     }
25     .content {
26       display: none;
27     }
28     .content.active {
29       display: block;
30     }
31   </style>
32 </head>
33 <body>
34   <div class="container">
35     <h1>Role-Based Access Control</h1>
36     <div class="role-select">
37       <label for="role">Select Role:</label>
38       <select id="role" onchange="renderContent()">
39         <option value="">--Select Role--</option>
40         <option value="admin">Admin</option>
41         <option value="editor">Editor</option>
42         <option value="user">User</option>
43       </select>
44     </div>
45     <div id="admin-content" class="content">
46       <h2>Admin Dashboard</h2>
47       <p>Welcome, Admin! You have full access to the system.</p>
48     </div>
49     <div id="editor-content" class="content">
50       <h2>Editor Dashboard</h2>
51       <p>Welcome, Editor! You can edit content but have limited access to settings.</p>
52     </div>
53     <div id="user-content" class="content">
54       <h2>User Dashboard</h2>
55       <p>Welcome, User! You can view content but cannot make changes.</p>
56     </div>
57   </div>
58
59   <script>
60     function renderContent() {
61       const role = document.getElementById('role').value;
62       document.querySelectorAll('.content').forEach(content => {
63         content.classList.remove('active');
64       });
65       if (role) {
66         document.getElementById(`${role}-content`).classList.add('active');
67       }
68     }
69   </script>
```


Output :



Task 5 – Multi-Language Website

Design a multilingual website.

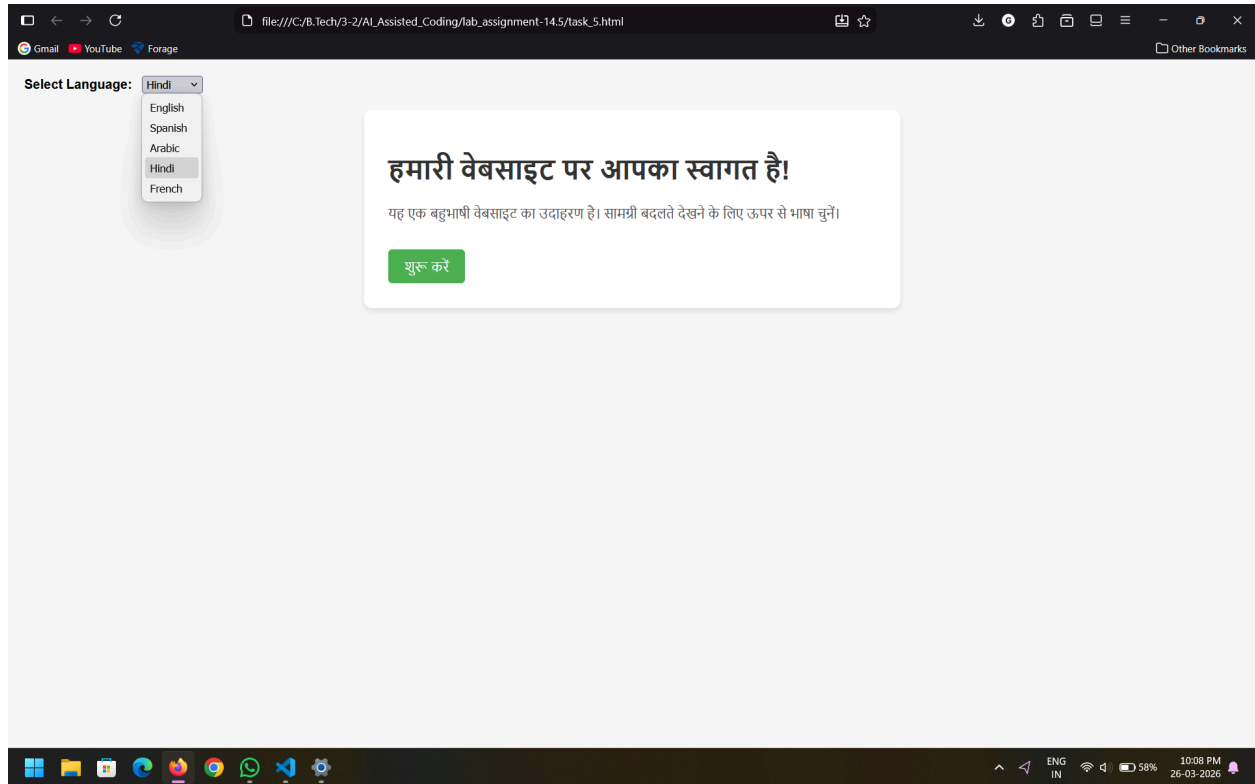
Requirements:

- Switch languages dynamically.
- Persist language preference.
- Use AI to:
 - o Structure translation files.
 - o Handle RTL/LTR layouts.
 - o Improve accessibility.

Copilot Suggested Code :

```
1  <!-- Design a multilingual website. -->
2  <!DOCTYPE html>
3  <html lang="en">
4
5  <head>
6    <title>Multi-Language Website</title>
7    <style>
8      body {
9        font-family: 'Arial', sans-serif;
10       margin: 0;
11       padding: 20px;
12       background-color: #f9f9f9;
13     }
14
15     .language-select {
16       margin-bottom: 20px;
17     }
18
19     .language-select label {
20       font-weight: bold;
21       margin-right: 8px;
22     }
23
24     .content {
25       max-width: 600px;
26       margin: 0 auto;
27       background-color: #fff;
28       padding: 30px;
29       border-radius: 10px;
30       box-shadow: 0 4px 8px #0000000.1;
31     }
32
33     .content h1 {
34       color: #333;
35     }
36
37     .content p {
38       color: #555;
39       line-height: 1.6;
40     }
41
42     .content button {
43       margin-top: 15px;
44       padding: 10px 20px;
45       background-color: #4CAF50;
46       color: white;
47       border: none;
48       border-radius: 5px;
49       cursor: pointer;
50       font-size: 16px;
51     }
52
53     .content button:hover {
54       background-color: #45a049;
55     }
56   </style>
57 </head>
58
59 <body>
60   <div class="language-select">
61     <label for="language">Select Language</label>
62     <select id="language" onchange="switchLanguage()">
63       <option value="en">English</option>
64       <option value="es">Spanish</option>
65       <option value="ar">Arabic</option>
66       <option value="hi">Hindi</option>
67       <option value="fr">French</option>
68     </select>
69   </div>
70   <div class="content" id="content">
71     <!-- Content will be dynamically updated based on language selection -->
72   </div>
73
74   <script>
75     const translations = {
76       en: {
77         welcome: "Welcome to our Website!",
78         description: "This is a multilingual website example. Select a language from the dropdown above to see the content change.",
79         btn: "Get Started"
80       },
81       es: {
82         welcome: "¡Bienvenido a nuestro sitio web!",
83         description: "Este es un ejemplo de sitio web multilingüe. Seleccione un idioma del menú desplegable para ver el contenido cambiar.",
84         btn: "Empezar"
85       },
86       ar: {
87         welcome: "مرحبًا بكم في موقعنا"،
88         description: "هذا مثال على موقع ويب متعدد اللغات. اختر لغة من القائمة المنسدلة لتغيير المحتوى.",
89         btn: "ابدأ"
90       },
91       hi: {
92         welcome: "स्वागत है आपका हमारे साइट पर!",
93         description: "यह एक बहुभाषी वेबसाइट का उदाहरण है। सम्पूर्ण बदलते देखने के लिए ऊपर से भाषा चुनें।",
94         btn: "शुरू करें"
95       },
96       fr: {
97         welcome: "Bienvenue sur notre site web!",
98         description: "Ceci est un exemple de site web multilingue. Sélectionnez une langue dans le menu déroulant pour voir le contenu changer.",
99         btn: "Commencer"
100      }
101    };
102
103    function switchLanguage() {
104      const language = document.getElementById('language').value;
105      localStorage.setItem('preferredLanguage', language);
106      renderContent(language);
107    }
108
109    function renderContent(language) {
110      const content = document.getElementById('content');
111      const t = translations[language];
112      content.innerHTML = `
113        <h1>${t.welcome}</h1>
114        <p>${t.description}</p>
115        <button>${t.btn}</button>
116      `;
117      document.body.dir = language === 'ar' ? 'rtl' : 'ltr';
118      document.documentElement.lang = language;
119    }
120
121    // Load preferred language on page load
122    const preferredLanguage = localStorage.getItem('preferredLanguage') || 'en';
123    document.getElementById('language').value = preferredLanguage;
124    renderContent(preferredLanguage);
125   </script>
126 </body>
127
```

Output :



Task 6 – Online Examination System

Create an online exam interface.

Requirements:

- Timer, questions, and submission.
- Prevent multiple submissions.
- Use AI to:
 - o Manage exam state.
 - o Improve exam UX.
 - o Ensure accessibility.

Copilot Suggested Code :

```
task_6.html > html > body
1 <!-- Task 6 - Online Examination System -->
2 <!DOCTYPE html>
3 <html lang="en">
4
5 <head>
6 <title>Online Examination System</title>
7 <style>
8   body {
9     font-family: Arial, sans-serif;
10    margin: 0;
11    padding: 20px;
12    background-color: #f9f9f9;
13  }
14
15  .exam-container {
16    max-width: 600px;
17    margin: 0 auto;
18    background-color: #fff;
19    padding: 20px;
20    border-radius: 10px;
21    box-shadow: 0 4px 8px rgba(0, 0, 0, 0.1);
22  }
23
24  h1 {
25    color: #333;
26  }
27
28  #timer {
29    font-size: 18px;
30    font-weight: bold;
31    color: #4CAF50;
32    margin-bottom: 20px;
33  }
34
35  #timer.warning {
36    color: orange;
37  }
38
39  #timer.danger {
40    color: red;
41  }
42
43  .question {
44    margin-bottom: 25px;
45    padding: 15px;
46    border: 1px solid #ddd;
47    border-radius: 8px;
48  }
49
50  .question p {
51    font-weight: bold;
52    margin-top: 0;
53  }
54
55  .options {
56    list-style-type: none;
57    padding: 0;
58  }
59
60  .options li {
61    margin-bottom: 10px;
62  }
63
64  .options label {
65    cursor: pointer;
66  }
67
68  .submit-btn {
69    padding: 10px 20px;
70    background-color: #4CAF50;
71    color: white;
72    border: none;
73    border-radius: 5px;
74    cursor: pointer;
75    font-size: 16px;
76  }
77
78  .submit-btn:hover {
79    background-color: #45a049;
80  }
81
82  .submit-btn.disabled {
83    background-color: #aaa;
84    cursor: not-allowed;
85  }
86
87  #result {
88    margin-top: 20px;
89    padding: 15px;
90    background-color: #e8f5e9;
91    border-radius: 8px;
92    display: none;
93  }
94 </style>
95 </head>
```

```

<body>
  <div class="exam-container">
    <h1>Online Examination</h1>
    <div id="timer">Time Remaining: 10:00</div>
    <div id="questions"></div>
    <button class="submit-btn" id="submitBtn" onClick="submitExam()">Submit Exam</button>
    <div id="result"></div>
  </div>

  <script>
    const examState = {
      timeRemaining: 600,
      questions: [
        { id: 1, text: "What is the capital of France?", options: ["Berlin", "Madrid", "Paris", "Rome"], answer: 2 },
        { id: 2, text: "What is 2 + 2?", options: ["3", "4", "5", "6"], answer: 1 },
        { id: 3, text: "Which planet is closest to the Sun?", options: ["Earth", "Venus", "Mercury", "Mars"], answer: 2 },
        { id: 4, text: "What is the boiling point of water (°C)?", options: ["90", "100", "110", "120"], answer: 1 },
        { id: 5, text: "Who developed Python programming language?", options: ["Dennis Ritchie", "Guido van Rossum", "James Gosling", "Bjarne Stroustrup"] },
      ],
      answers: {},
      submitted: false
    };

    let timerInterval;

    function startTimer() {
      const timerElement = document.getElementById('timer');
      timerInterval = setInterval(() => {
        if (examState.timeRemaining <= 0) {
          clearInterval(timerInterval);
          submitExam();
        } else {
          examState.timeRemaining--;
          const minutes = Math.floor(examState.timeRemaining / 60);
          const seconds = examState.timeRemaining % 60;
          timerElement.textContent = `Time Remaining: ${minutes}:${seconds < 10 ? '0' : ''}${seconds}`;

          if (examState.timeRemaining <= 60) {
            timerElement.className = 'danger';
          } else if (examState.timeRemaining <= 180) {
            timerElement.className = 'warning';
          }
        }
      }, 1000);
    }

    function renderQuestions() {
      const questionsContainer = document.getElementById('questions');
      examState.questions.forEach((question, qIndex) => {
        const questionElement = document.createElement('div');
        questionElement.className = 'question';
        questionElement.setAttribute('role', 'group');
        questionElement.setAttribute('aria-labelledby', `q${question.id}`);
        questionElement.innerHTML = `
        <p id="q${question.id}">${qIndex + 1}. ${question.text}</p>
        <ul class="options">
          ${question.options.map((option, index) => `
            <li>
              <label>
                <input type="radio" name="question-${question.id}" value="${index}"
                  onchange="selectAnswer(${question.id}, ${index})"
                  aria-label="${option}">
                ${option}
              </label>
            </li>
          `).join('')}
        </ul>
        `;
        questionsContainer.appendChild(questionElement);
      });
    }

    function selectAnswer(questionId, optionIndex) {
      examState.answers[questionId] = optionIndex;
    }

    function submitExam() {
      if (examState.submitted) {
        alert("You have already submitted the exam!");
        return;
      }

      clearInterval(timerInterval);
      examState.submitted = true;

      document.querySelectorAll('input[type="radio"]').forEach(input => input.disabled = true);
      document.getElementById('submitBtn').disabled = true;
      document.getElementById('timer').textContent = "Exam Submitted!";

      // Calculate score
      let score = 0;
      examState.questions.forEach(q => {
        if (examState.answers[q.id] === q.answer) score++;
      });

      const total = examState.questions.length;
      const percent = Math.round((score / total) * 100);

      const resultDiv = document.getElementById('result');
      resultDiv.style.display = 'block';
      resultDiv.innerHTML = `
      <h2>Exam Results</h2>
      <p>You scored <strong>${score} / ${total}</strong> (${percent}%)</p>
      <p>${percent >= 60 ? 'Congratulations! You passed!' : 'You did not pass. Better luck next time!'}</p>
      `;
      resultDiv.scrollIntoView({ behavior: 'smooth' });
    }

    renderQuestions();
    startTimer();
  </script>
</body>
</html>

```

Output :

The screenshot shows a web browser window with a dark theme. The address bar displays the file path: `file:///C:/B.Tech/3-2/AI_Assisted_Coding/lab_as`. Below the address bar, there are bookmarks for Gmail, YouTube, and Forage, and a link to Other Bookmarks. The main content area is titled "Online Examination" in bold black text. Below the title, a green text label indicates "Time Remaining: 9:40". The examination consists of five multiple-choice questions, each in a separate box:

- 1. What is the capital of France?**
 - ☐ Berlin
 - ☐ Madrid
 - ☐ Paris
 - ☐ Rome
- 2. What is 2 + 2?**
 - ☐ 3
 - ☐ 4
 - ☐ 5
 - ☐ 6
- 3. Which planet is closest to the Sun?**
 - ☐ Earth
 - ☐ Venus
 - ☐ Mercury
 - ☐ Mars
- 4. What is the boiling point of water (°C)?**
 - ☐ 90
 - ☐ 100
 - ☐ 110
 - ☐ 120
- 5. Who developed Python programming language?**
 - ☐ Dennis Ritchie
 - ☐ Guido van Rossum
 - ☐ James Gosling
 - ☐ Bjarne Stroustrup

At the bottom of the examination area, there is a green button labeled "Submit Exam". The Windows taskbar is visible at the bottom of the screen, showing various application icons and system status information: ENG IN, 54% battery, and the date/time 10:20 PM 26-03-2026.